

Data Cleaning

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Description

This cleaning file evolved over the course of the project but the final goal for cleaning the data was to provide a sample of games that are Steam-only or offered on both Steam and Epic. The sample of games are limited by release date (2013 - 2023) and each game's data is cut off in late 2024. Although only Steam's pricing data is used in the project, Epic data is included to define the sample of games that are offered on both platforms.

This cleaning file could easily be repurposed to engineer a dataset with multiple vendors and the pricing data from those platforms, but for the sake of this project, I only required Steam data.

Data Sources

The two primary datasets used in this file (pricing and metadata) were collected from the IsThereAnyDeal API. CPI data was collected from the U.S. Bureau of Labor Statistics.

Steps of Cleaning

- Load Libraries and Datasets
- Clean Pricing Data
 - Filter for Steam and Epic observations
 - Filter out data past December 6, 2024 to give the latest games at least one year of data
 - Filter out excluded games
- Clean Metadata
- Filter for games released 5 years before and after Epic's entry
- Reformat genre data into 14 genre dummy variables
- Prep CPI data and merge with pricing data for inflation-adjusted prices
- Merge pricing data and metadata
- Clean main dataset by dropping unused variables and outlier observations
- Reformat the main dataset to a daily format

- Fill all unobserved days with the previous observation's values
- Extend each game-shop combination to the last sold date
- Merge datasets
- Make final adjustments (Sale and Days Since Release variables)
- Export the data

Notes / Important Information

- 2015 is used as the base year for CPI data
 - Free games were dropped due to lack of price variation
 - Epic Only games were not studied because of the small sample size
 - One potentially controversial decision I made when cleaning the data was excluding all observations where a game's regular (undiscounted) price exceeded \$60 for titles released before 2021 or \$70 for titles released in 2021 or later. The API tracks all price changes, even misinputs, so most / all extreme values were clearly the result of seller pricing errors rather than actual price changes. The price caps I used were based on typical price ceilings in this market.
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Libraries and Data

```
# The only library needed here is tidyverse
library(tidyverse)
library(lubridate)

# Load data
prices <- read_csv("Data/pricing_history.csv")
metadata <- read_csv("Data/games_metadata.csv")
cpi <- read_csv("Data/CPI.csv")
```

Data Cleaning

Initial Cleaning of Pricing Data

```
prices <- prices %>%
  # Filter for Steam and Epic
  filter(str_detect(shop_name, "Steam") | str_detect(shop_name, "Epic")) %>%
  # Rename all instances of Epic Game Store to Epic (Personal Preference)
  mutate(shop_name = if_else(str_detect(shop_name, "Epic Game Store"), "Epic", shop_name)) %>%
  # Delete data past December 6, 2024
  filter(timestamp <= as.Date("2024-12-06")) %>%
  # Filter for USD only
  filter(currency == "USD")
```

All free games were excluded so that the lack of price variation would not break later statistical tests. Other

games were excluded due to not being the full version of games (Ex: separately listed DLC, Demos) or not being games at all (Wallpaper Engine). Additionally, some games switched to a free format during the sample period. Those games had their data cutoff after they switched to being free.

```
# List of Excluded Games
excluded_games <- c("Brawlhalla", "War Thunder", "Quake Champions", "Destiny 2",
  "Team Fortress 2", "Path of Exile", "Dota 2", "Scorn",
  "Counter-Strike: Global Offensive", "Crusader Kings II",
  "Life is Strange - Episode 1", "Apex Legends", "Warframe",
  "Life is Strange 2", "Cyberpunk 2077: Phantom Liberty",
  "Borderlands 3", "Sid Meier's Civilization® VI", "HITMAN",
  "Tiny Tina's Wonderlands", "Hitman Sniper Challenge",
  "For Honor - Public Test", "Hitman 2 Prologue", "LAB",
  "Medal of Honor(TM) Multiplayer", "Black Desert Online",
  "Counter-Strike 2", "Minion Masters", "Kingdom: Classic",
  "Manor Lords", "Wallpaper Engine", "Grand Theft Auto V Enhanced",
  "The Expendabros", "Z1 Battle Royale", "PUBG: BATTLEGROUNDS",
  "Tell Me Why: Chapter 1", "Jagged Alliance 1: Gold Edition",
  "A Space For The Unbound", "WARSAW RISING: City of Heroes",
  "Tell Me Why")

# Filter out all excluded games
prices <- prices %>%
  # Filter based on the excluded games list
  filter(!game_title %in% excluded_games) %>%
  # Exclude The Sims 4 after October 18, 2022
  filter(!(game_title == "The Sims 4" & timestamp >= as.Date("2022-10-18"))) %>%
  # Exclude Rocket League after September 23, 2020
  filter(!(game_title == "Rocket League" & timestamp >= as.Date("2020-09-23"))) %>%
  # Excluded Totally Accurate Battlegrounds after April 1, 2021
  filter(
    !(game_title == "Totally Accurate Battlegrounds" & timestamp >= as.Date("2021-04-01"))
  )
```

Clean Metadata

I opted to restrict the sample to games released 5 years before and after Epic launched their platform.

```
# Filter metadata by release date
metadata_filtered <- metadata %>%
  filter(release_date >= as.Date("2013-12-06") &
    release_date <= as.Date("2023-12-06"))
```

Gather Genres from Tags

In order to create functioning genre variables, I had to take all genre tags from the raw data and cluster them into broader genres. After reviewing the tendency of all genre tags, I constructed 14 unique genres all functioning as indicator variables that check if the game's list of genre tags include any of the related keywords.

```
# List each genre tag for its matching genre
RPG_tags <- c("RPG", "JRPG", "CRPG", "Action RPG", "Party-Based RPG", "Tactical RPG",
  "Dungeon Crawler", "Choices Matter", "Character Customization", "Loot")
Action_tags <- c("Action", "Action-Adventure", "Hack and Slash", "Souls-like",
  "Beat 'em up", "Fighting", "Character Action Game", "Combat", "Spectacle fighter",
  "Martial Arts", "Swordplay")
```

```

Adventure_tags <- c( "Adventure", "Exploration", "Walking Simulator", "Point & Click",
  "Mystery", "Detective", "Interactive Fiction")
Shooter_tags <- c("FPS", "Shooter", "First-Person Shooter", "Third-Person Shooter",
  "Top-Down Shooter", "Twin Stick Shooter", "Looter Shooter", "Hero Shooter",
  "Arena Shooter", "Boomer Shooter", "Extraction Shooter", "Shoot 'Em Up",
  "Bullet Hell", "Sniper")
Strategy_tags <- c("Strategy", "RTS", "4X", "Turn-Based Strategy", "Turn-Based Tactics",
  "Tactical", "Tower Defense", "Grand Strategy", "Real Time Tactics", "MOBA")
Simulation_tags <- c("Simulation", "Life Sim", "City Builder", "Colony Sim",
  "Automobile Sim", "Farming Sim", "Political Sim", "Management", "Resource Management",
  "Economy", "God Game", "Outbreak Sim", "Medical Sim")
Platformer_tags <- c("Platformer", "2D Platformer", "3D Platformer", "Puzzle Platformer",
  "Precision Platformer", "Metroidvania", "Side Scroller", "Runner", "Parkour")
Puzzle_tags <- c("Puzzle", "Logic", "Hidden Object", "Escape Room")
Horror_tags <- c("Horror", "Survival Horror", "Psychological Horror", "Psychological",
  "Lovecraftian", "Supernatural", "Gore", "Dark")
OpenWorld_tags <- c("Open World", "Sandbox", "Open World Survival Craft")
Survival_tags <- c("Survival", "Crafting", "Base Building", "Building",
  "Mining", "Automation")
Roguelike_tags <- c("Roguelike", "Roguelite", "Action Roguelike",
  "Roguelike Deckbuilder", "Roguevania", "Procedural Generation", "Perma Death")
Multiplayer_tags <- c("Multiplayer", "Online Co-Op", "Local Co-Op", "Local Multiplayer",
  "PvP", "PvE", "Team-Based", "Competitive", "Battle Royale")
Narrative_tags <- c("Story Rich", "Narrative", "Multiple Endings",
  "Choices Matter", "Visual Novel", "Cinematic", "Lore-Rich", "Emotional")

# Create genre dummies based on tags
metadata_dummies <- metadata_filtered %>%
  separate_rows(tags, sep = ",") %>%
  mutate(tags = str_trim(tags)) %>%
  mutate(
    RPG = if_else(tags %in% RPG_tags, 1, 0),
    Action = if_else(tags %in% Action_tags, 1, 0),
    Adventure = if_else(tags %in% Adventure_tags, 1, 0),
    Shooter = if_else(tags %in% Shooter_tags, 1, 0),
    Strategy = if_else(tags %in% Strategy_tags, 1, 0),
    Simulation = if_else(tags %in% Simulation_tags, 1, 0),
    Platformer = if_else(tags %in% Platformer_tags, 1, 0),
    Puzzle = if_else(tags %in% Puzzle_tags, 1, 0),
    Horror = if_else(tags %in% Horror_tags, 1, 0),
    OpenWorld = if_else(tags %in% OpenWorld_tags, 1, 0),
    Survival = if_else(tags %in% Survival_tags, 1, 0),
    Roguelike = if_else(tags %in% Roguelike_tags, 1, 0),
    Multiplayer = if_else(tags %in% Multiplayer_tags, 1, 0),
    Narrative = if_else(tags %in% Narrative_tags, 1, 0)
  ) %>%
  group_by(game_id) %>%
  summarise(
    title = first(title),
    release_date = first(release_date),
    publishers = first(publishers),
    RPG = max(RPG),
    Action = max(Action),

```

```

Adventure = max(Adventure),
Shooter = max(Shooter),
Strategy = max(Strategy),
Simulation = max(Simulation),
Platformer = max(Platformer),
Puzzle = max(Puzzle),
Horror = max(Horror),
OpenWorld = max(OpenWorld),
Survival = max(Survival),
Roguelike = max(Roguelike),
Multiplayer = max(Multiplayer),
Narrative = max(Narrative)
)

```

Prepare CPI Data

```

# Convert CPI to datetime monthly value
cpi_monthly_clean <- cpi %>%
  mutate(
    # Create a proper date for the CPI month
    cpi_month = ymd(paste(Year, Period, "01")),
    cpi = as.numeric(Value)
  ) %>%
  select(cpi_month, cpi)

```

```

## Warning: There was 1 warning in `mutate()`.
## i In argument: `cpi = as.numeric(Value)`.
## Caused by warning:
## ! NAs introduced by coercion

```

```

# Get base year value
cpi_2015 <- cpi_monthly_clean %>%
  filter(year(cpi_month) == 2015) %>%
  summarise(cpi_2015 = mean(cpi)) %>%
  pull(cpi_2015)

```

Apply Inflation Adjustment to Prices

```

prices <- prices %>%
  # Get corresponding month to pricing data points
  mutate(
    date = as.Date(timestamp),
    cpi_month = floor_date(date, "month")
  ) %>%
  # Merge with CPI data
  left_join(
    cpi_monthly_clean,
    by = "cpi_month"
  ) %>%
  # Calculate real price
  mutate(
    real_price = price * (cpi_2015 / cpi),
    real_regular_price = regular_price * (cpi_2015 / cpi)
  )

```

)

Combine Metadata and Pricing Data

```
# Merge metadata with pricing data
main_df <- prices %>%
  semi_join(metadata_dummies, by = "game_id") %>%
  left_join(metadata_dummies, by = "game_id")
```

Clean Final Dataset

```
main_df <- main_df %>%
  # Filter for games with at least 10 price points
  group_by(game_title) %>%
  filter(n() >= 10) %>%
  ungroup() %>%
  # Drop unused columns
  select(-c(cpi_month, cpi, title, game_id, currency, timestamp, shop_id))

# Exclude observations where regular price > $60 (pre-2021) or > $70 (post-2021)
main_df <- main_df %>%
  filter(
    (release_date < as.Date("2021-01-01") & regular_price <= 60) |
    (release_date >= as.Date("2021-01-01") & regular_price <= 70)
  )

# Create steam dummy variable
main_df <- main_df %>%
  mutate(steam = if_else(str_detect(shop_name, "Steam"), 1, 0))
```

Make Daily Data

```
# Calculate cutoff date per shop (Last observed date per shop, used as the
# common end point for all games)
shop_cutoffs <- main_df %>%
  group_by(shop_name) %>%
  summarise(cutoff_date = max(date), .groups = "drop")
```

```
# Step 1: Daily Expansion
main_df_daily <- main_df %>%
  group_by(game_title, shop_name) %>%
  complete(
    date = seq.Date(min(date), max(date), by = "day")
  ) %>%
  fill(
    game_title, release_date, publishers,
    RPG:Narrative,
    price, regular_price,
    real_price, real_regular_price,
    discount_percent, steam,
    .direction = "down"
  ) %>%
```

```

ungroup()

# Step 2: For each game-shop, extend from its last date to the shop cutoff
extension <- main_df_daily %>%
  left_join(shop_cutoffs, by = "shop_name") %>%
  group_by(game_title, shop_name) %>%
  filter(date == max(date) & date < cutoff_date) %>%
  slice(1) %>%
  group_modify(~ {
    extra_dates <- seq.Date(.x$date + 1, .x$cutoff_date, by = "day")
    .x[rep(1, length(extra_dates)), ] %>% mutate(date = extra_dates)
  }) %>%
  ungroup() %>%
  select(-cutoff_date)

# Step 3: Bind together
main_df_daily <- bind_rows(main_df_daily, extension) %>%
  arrange(game_title, shop_name, date)

main_df_daily <- main_df_daily %>%
  # Create "sale" variable (if discount_percent > 0, then sale = 1, else 0)
  mutate(sale = if_else(discount_percent > 0, 1, 0)) %>%
  # Create days since release variable
  mutate(days_since_release = as.numeric(date - release_date))

```

Export Cleaned Dataset

```
write_csv(main_df_daily, "Data/cleaned_data.csv")
```

Extra Information and Code

Because this data cleaning file evolved so much over time to fit the changing demands of the project, there were many code artifacts left behind. I cleaned these artifacts from this version to make the file cleaner and more readable, but I have provided some of the useful code segments here for repurposing the file for a more extensive final dataset.

List of Resellers

```

resellers <- c("Humble Store", "Fanatical", "GreenManGaming", "IndieGala Store",
  "GamesPlanet US", "GameBillet", "WinGameStore", "GamesPlanet UK",
  "GamersGate", "GamesPlanet FR", "GamesPlanet DE", "DLGamer",
  "AllYouPlay", "Newegg", "Dreamgame", "Nuuvem", "2game", "Gamesload",
  "Noctre", "JoyBuggy", "GamesRepublic", "GameStop PC", "Savemi",
  "Razer Game Store", "Direct2Drive", "eTail.Market", "MacGameStore",
  "SilaGames", "Chrono.gg", "PlanetPlay", "Gamer Thor", "Playfield",
  "Digital Download", "Gamesrocket", "GameFly", "GetGames",
  "Humble Widgets", "Playsum", "GameTap", "FunStock Digital",
  "Imperial Games", "Gemly", "Desura", "IndieGameStand", "Discord Store",
  "Last Best Offer", "ShinyLoot", "Coinplay.io", "Playism", "Oculus Store",
  "Gameolith", "FireFlower", "Game Jolt", "Adventure Shop", "BI Store")

```

Epic-Only Games

For those interested, this is the list of Epic-only games that I could find from my original dataset: Lego 2k Drive, Borderlands 3, Sifu, Marvel's Midnight Suns, Civilization VI, Maneater, Chivalry 2, Crime Boss, Tchia, Scorn, and Tiny Tina

Clean Publishers

For those interested in cleaning the developer data included in the raw data, this is the progress I made before opting to not use developers in the final version of the project. This list should cover all games included in my sample. The ordering does matter here since Steam lists all developers responsible for the game when I was only focused on the primary developer.

```
# clean publishers column
main_df <- main_df %>%
  mutate(publishers = case_when(
    str_detect(publishers, "Remedy Entertainment") ~ "Remedy Entertainment",
    str_detect(publishers, "VOID Interactive") ~ "VOID Interactive",
    str_detect(publishers, "The Fun Pimps") ~ "The Fun Pimps",
    str_detect(publishers, "Ubisoft") ~ "Ubisoft",
    str_detect(publishers, "Electronic Arts") ~ "EA",
    str_detect(publishers, "Square Enix") ~ "Square Enix",
    str_detect(publishers, "Eidos") ~ "Square Enix",
    str_detect(publishers, "PlayStation") ~ "Playstation",
    str_detect(publishers, "Capcom") ~ "Capcom",
    str_detect(publishers, "Sega") ~ "Sega",
    str_detect(publishers, "SEGA") ~ "Sega",
    str_detect(publishers, "Bethesda") ~ "Bethesda",
    str_detect(publishers, "FromSoftware") ~ "FromSoftware",
    str_detect(publishers, "Activision") ~ "Activision",
    str_detect(publishers, "Xbox Game Studios") ~ "Xbox Game Studios",
    str_detect(publishers, "THQ") ~ "THQ",
    str_detect(publishers, "Focus Entertainment") ~ "Focus Entertainment",
    str_detect(publishers, "Larian") ~ "Larian",
    str_detect(publishers, "11 bit studios") ~ "11 bit studios",
    str_detect(publishers, "Devolver") ~ "Devolver Digital",
    str_detect(publishers, "tinyBuild") ~ "tinyBuild",
    str_detect(publishers, "Humble Games") ~ "Humble Games",
    str_detect(publishers, "Klei Entertainment") ~ "Klei Entertainment",
    str_detect(publishers, "Galactic Cafe") ~ "Galactic Cafe",
    str_detect(publishers, "Everything Unlimited Ltd.") ~ "Everything Unlimited Ltd.",
    str_detect(publishers, "Paradox Interactive") ~ "Paradox Interactive",
    str_detect(publishers, "BANDAI NAMCO Entertainment") ~ "BANDAI NAMCO Entertainment",
    str_detect(publishers, "Ninja Theory") ~ "Ninja Theory",
    str_detect(publishers, "StudioMDHR") ~ "StudioMDHR",
    str_detect(publishers, "Annapurna") ~ "Annapurna Interactive",
    str_detect(publishers, "Coffee Stain") ~ "Coffee Stain Publishing",
    str_detect(publishers, "Secret Mode") ~ "Secret Mode",
    str_detect(publishers, "Curve") ~ "Curve Digital",
    str_detect(publishers, "RyseUp") ~ "RyseUp Studios",
    str_detect(publishers, "Starbreeze") ~ "Starbreeze Entertainment",
    str_detect(publishers, "Prime Matter") ~ "Plaion",
    str_detect(publishers, "Tripwire") ~ "Tripwire Interactive",
    str_detect(publishers, "Lince Works") ~ "Lince Works",
    str_detect(publishers, "Deep Silver") ~ "Plaion",
```

```

str_detect(publishers, "Unknown") ~ "Unknown Worlds Entertainment",
str_detect(publishers, "Rockstar Games") ~ "Rockstar Games",
str_detect(publishers, "Private Division") ~ "Private Division",
str_detect(publishers, "2K") ~ "2K Games",
str_detect(publishers, "Hello Games") ~ "Hello Games",
str_detect(publishers, "Konami") ~ "Konami Digital Entertainment",
str_detect(publishers, "KOJIMA") ~ "KOJIMA PRODUCTIONS",
str_detect(publishers, "Remedy Entertainment") ~ "Remedy Entertainment",
str_detect(publishers, "Thing Trunk") ~ "Thing Trunk",
str_detect(publishers, "Raiser Games") ~ "Raiser Games",
str_detect(publishers, "Duoyi Games") ~ "Duoyi Games",
str_detect(publishers, "Neocore Games") ~ "Neocore Games",
str_detect(publishers, "Kunos Simulazioni") ~ "Kunos Simulazioni",
str_detect(publishers, "505") ~ "505 Games",
str_detect(publishers, "New World Interactive") ~ "New World Interactive",
str_detect(publishers, "Frontier") ~ "Frontier Developments",
str_detect(publishers, "Tuxedo") ~ "Tuxedo Labs",
str_detect(publishers, "Beamdog") ~ "Beamdog",
str_detect(publishers, "Quantic") ~ "Quantic Dream",
str_detect(publishers, "Aspyr") ~ "Aspyr",
str_detect(publishers, "CD PROJEKT RED") ~ "CD PROJEKT RED",
str_detect(publishers, "Daedalic") ~ "Daedalic Entertainment",
str_detect(publishers, "Fellow") ~ "Fellow Traveller",
str_detect(publishers, "Fatshark") ~ "Fatshark",
str_detect(publishers, "Monomi") ~ "Monomi Park",
str_detect(publishers, "Thunder Lotus") ~ "Thunder Lotus",
str_detect(publishers, "The Game Bakers") ~ "The Game Bakers",
str_detect(publishers, "Behold Studios") ~ "Behold Studios",
str_detect(publishers, "Wired Productions") ~ "Wired Productions",
str_detect(publishers, "Fishing Cactus") ~ "Fishing Cactus",
str_detect(publishers, "Frozenbyte") ~ "Frozenbyte",
str_detect(publishers, "Spiderling") ~ "Spiderling Studios",
str_detect(publishers, "Telltale") ~ "Telltale Games",
str_detect(publishers, "Thunderful") ~ "Thunderful Publishing",
str_detect(publishers, "Operators") ~ "Games Operators",
str_detect(publishers, "Bedtime") ~ "Bedtime Digital Games",
str_detect(publishers, "Dear") ~ "Dear Villagers",
str_detect(publishers, "SUPERHOT") ~ "SUPERHOT Team",
str_detect(publishers, "IMGN") ~ "IMGN",
str_detect(publishers, "WayForward") ~ "WayForward",
str_detect(publishers, "Nacon") ~ "Nacon",
str_detect(publishers, "Playtonic Friends") ~ "Playtonic Friends",
str_detect(publishers, "Team17") ~ "Team17",
str_detect(publishers, "Warner Bros") ~ "WB Games",
str_detect(publishers, "IO Interactive") ~ "IO Interactive",
str_detect(publishers, "Spearhead Games") ~ "Spearhead Games",
str_detect(publishers, "3909") ~ "3909",
str_detect(publishers, "tobyfox") ~ "tobyfox",
str_detect(publishers, "E-Line Media") ~ "E-Line Media",
str_detect(publishers, "Hooded Horse") ~ "Hooded Horse",
str_detect(publishers, "Hidden Folks") ~ "Hidden Folks",
str_detect(publishers, "Alawar") ~ "Alawar",
str_detect(publishers, "Almost Human Games") ~ "Almost Human Games",

```

```

str_detect(publishers, "Behaviour") ~ "Behaviour Interactive",
str_detect(publishers, "Black Tabby Games") ~ "Black Tabby Games",

str_detect(publishers, "Playstack") ~ "Playstack",
str_detect(publishers, "Bloofer Team") ~ "Bloofer Team",
str_detect(publishers, "Bohemia Interactive") ~ "Bohemia Interactive",
str_detect(publishers, "Chucklefish") ~ "Chucklefish",
str_detect(publishers, "D-Pad Studio") ~ "D-Pad Studios",
str_detect(publishers, "Data Realms") ~ "Data Realms",
str_detect(publishers, "Digerati") ~ "Digerati Distribution",
str_detect(publishers, "Eleon Game Studios") ~ "Eleon Game Studios",
str_detect(publishers, "Flippfly") ~ "Flippfly",
str_detect(publishers, "Giant Army") ~ "Giant Army",
str_detect(publishers, "Greenheart Games") ~ "Greenheart Games",
str_detect(publishers, "Hinterland Studio") ~ "Hinterland Studio",
str_detect(publishers, "Jackbox Games") ~ "Jackbox Games",
str_detect(publishers, "KOEI TECMO") ~ "Koei Tecmo",
str_detect(publishers, "Kalypso") ~ "Kalypso Media",
str_detect(publishers, "Keen Games") ~ "Keen Games",
str_detect(publishers, "Killhouse Games") ~ "Killhouse Games",
str_detect(publishers, "Kitfox Games") ~ "Kitfox Games",
str_detect(publishers, "Landfall") ~ "Landfall Games",
str_detect(publishers, "Lion Shield") ~ "Lion Shield",
str_detect(publishers, "Lo-Fi Games|Lo Fi Games") ~ "Lo-Fi Games",
str_detect(publishers, "Mechanistry") ~ "Mechanistry",
str_detect(publishers, "Milkstone") ~ "Milkstone Studios",
str_detect(publishers, "Nicalis") ~ "Nicalis",
str_detect(publishers, "OMOCAT") ~ "OMOCAT",
str_detect(publishers, "Oddworld Inhabitants") ~ "Oddworld Inhabitants",
str_detect(publishers, "Offworld Industries") ~ "Offworld Industries",
str_detect(publishers, "Pillow Castle") ~ "Pillow Castle",
str_detect(publishers, "Polytron") ~ "Polytron Corporation",
str_detect(publishers, "Rebellion") ~ "Rebellion Developments",
str_detect(publishers, "Red Candle Games") ~ "Red Candle Games",
str_detect(publishers, "Red Hook Studios") ~ "Red Hook Studios",
str_detect(publishers, "Robot Entertainment") ~ "Robot Entertainment",
str_detect(publishers, "Rockfish Games") ~ "Rockfish Games",
str_detect(publishers, "SCS Software") ~ "SCS Software",
str_detect(publishers, "Shining Rock Software") ~ "Shining Rock Software",
str_detect(publishers, "Shiro Games") ~ "Shiro Games",
str_detect(publishers, "Slitherine") ~ "Slitherine Ltd.",
str_detect(publishers, "Spike Chunsoft") ~ "Spike Chunsoft",
str_detect(publishers, "Squanch Games") ~ "Squanch Games",
str_detect(publishers, "Stardock") ~ "Stardock Entertainment",
str_detect(publishers, "Studio Wildcard") ~ "Studio Wildcard",
str_detect(publishers, "Stunlock Studios") ~ "Stunlock Studios",
str_detect(publishers, "System Era Softworks") ~ "System Era Softworks",
str_detect(publishers, "Wube Software") ~ "Wube Software",

# Publisher is one of several listed - main publisher identified
str_detect(publishers, "Versus Evil|Stoic") ~ "Versus Evil",
str_detect(publishers, "Boneloaf|Double Fine") ~ "Double Fine Productions",
str_detect(publishers, "Campo Santo|Panic") ~ "Campo Santo",

```

```

str_detect(publishers, "Codemasters") ~ "Codemasters",
str_detect(publishers, "Crystal Dynamics") ~ "Crystal Dynamics",
str_detect(publishers, "Deck13") ~ "Deck13",
str_detect(publishers, "Double Damage") ~ "Double Damage Games",
str_detect(publishers, "DrinkBox Studios|Drinkbox Studios") ~ "DrinkBox Studios",
str_detect(publishers, "Fireshine Games|Sold Out") ~ "Fireshine Games",
str_detect(publishers, "Freebird Games") ~ "Freebird Games",
str_detect(publishers, "Frogwares") ~ "Frogwares",
str_detect(publishers, "Frozen District") ~ "Frozen District",
str_detect(publishers, "Fruitbat Factory") ~ "Fruitbat Factory",
str_detect(publishers, "Fullbright") ~ "Fullbright",
str_detect(publishers, "Gamera Game") ~ "Gamera Game",
str_detect(publishers, "Gearbox Publishing") ~ "Gearbox Publishing",
str_detect(publishers, "Green Man Loaded|10 Chambers") ~ "10 Chambers",
str_detect(publishers, "Green Man Loaded|VOID Interactive") ~ "VOID Interactive",
str_detect(publishers, "Half Mermaid") ~ "Half Mermaid Productions",
str_detect(publishers, "HandyGames|Handy Games") ~ "HandyGames",
str_detect(publishers, "Hopoo Games") ~ "Hopoo Games",
str_detect(publishers, "ION LANDS") ~ "ION LANDS",
str_detect(publishers, "Kepler Interactive") ~ "Kepler Interactive",
str_detect(publishers, "META Publishing|Owlcat Games") ~ "Owlcat Games",
str_detect(publishers, "Matt Makes Games|Maddy Makes Games") ~ "Maddy Makes Games",
str_detect(publishers, "Microsoft.*Crytek|Crytek.*Microsoft") ~ "Xbox Game Studios",
str_detect(publishers, "NEXON|MINTROCKET") ~ "MINTROCKET",
str_detect(publishers, "New Game order|ComonGames") ~ "META Publishing",
str_detect(publishers, "Night School Studio") ~ "Night School Studio",
str_detect(publishers, "Nightdive Studios") ~ "Nightdive Studios",
str_detect(publishers, "Nihon Falcom|XSEED") ~ "XSEED Games",
str_detect(publishers, "Raw Fury") ~ "Raw Fury",
str_detect(publishers, "Serenity Forge") ~ "Serenity Forge",
str_detect(publishers, "ZA/UM|Disco Elysium") ~ "ZA/UM",
str_detect(publishers, "Playdigious") ~ "Playdigious",
str_detect(publishers, "Techland") ~ "Techland",
str_detect(publishers, "Team Reptile") ~ "Team Reptile",
str_detect(publishers, "Toplitz Productions") ~ "Toplitz Productions",
str_detect(publishers, "The Irregular Corporation") ~ "The Irregular Corporation",
str_detect(publishers, "Tequila Works") ~ "Tequila Works",
str_detect(publishers, "Thekla Inc.") ~ "Thekla Inc.",
str_detect(publishers, "Ysbryd Games") ~ "Ysbryd Games",
str_detect(publishers, "Worldwalker Games") ~ "Worldwalker Games",
TRUE ~ publishers
))

```